Executable Evidence Management for CAD-Guided Aircraft Inspection

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Abstract

Engineering changes alter the evidence supporting inspection outputs. We develop executable operating contracts that connect CAD-guided aircraft inspection requirements to versioned numerical producers and calibrated statistical statements. A typed representation and unit-aware compiler link task events, provider dependencies, numerical sources and evidence. Requirement containment supports reuse, requested-function invariance supports selective renewal, and persistent execution history governs issuance. The evaluation independently generates 384 metallic CAD-surface scenes and executes six imaging/estimation configurations, totaling 2,304 producer runs. Pixel-derived fastener correspondences refine pose and support physical image-to-CAD scoring. A decoder-coordinate update leaves a median reprojection residual of 0.261 pixel while increasing median mapping error to 8.217 mm, exposing the need to bind evidence to the effective producer. At risk allowance 0.15, reuse with exact-event fallback preserves all 44 directly supported contexts. It constructs 25 proofs rather than 42 for dependency-aware exact renewal, reducing complete-catalogue record evaluations from 8,064 to 4,800, or 40.48%. Requested dependency projection also reduces full image-pipeline evaluations by 25% across individual cold update/request cases. The framework converts requirement, component and execution changes into traceable evidence-retention, renewal and issuance decisions.

Keywords: Executable operating contracts, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

An aircraft inspection image becomes engineering information when it can be located on the aircraft model and its quality can be related to the requested task. Acquisition settings, feature extraction and pose estimation jointly determine that relationship. A larger region of interest can reveal more surface detail, a longer exposure can change the localization of a fastener center, and a decoder update can alter the coordinate convention used by the downstream estimator. The output can remain visually plausible while the evidence supporting its use has changed.

This creates a recurring engineering decision: which previous quality statements remain applicable after a requirement or component changes? Recomputing every statement discards useful evidence. Retaining every cached statement can attach a valid bound to a different producer. The required operation is selective: identify the quantities used by the current request, retain evidence whose event and producer remain compatible, and renew the affected portion.

Engineering knowledge models connect requirements, decisions and dependencies [1, 2, 3]. Risk-controlled selection connects a statistical statement to a specified effective program [4]. Runtime assurance connects monitored assumptions to permissible execution [5]. For inspection, these relationships must meet at the output actually produced. A change-impact relation becomes actionable when it identifies both the certified failure event and the numerical version used in the image-to-CAD mapping.

We develop executable operating contracts for evidence management in CAD-guided aircraft inspection. Unit-typed requirements compile into events and provider projections. Event containment supports a changed requirement; requested-function invariance supports an unchanged quantity after a component update.

Versioned execution records connect the selected statement to the output. Figure 1 documents the project setting, and Fig. 2 gives the executable architecture.

The study makes three contributions. First, it provides a typed representation and compiler linking requirements, producers, numerical versions and calibrated evidence. Second, it composes admission, transfer and persistent-history rules into a reuse-first workflow with exact-event fallback. Third, it evaluates the workflow on independently generated CAD-surface scenes, actual pixel extraction and pose refinement, and matched evidence-renewal alternatives. The resulting comparison preserves all 44 directly supported contexts while reducing complete-catalogue record evaluations from 8,064 to 4,800 relative to dependency-aware exact renewal. A coordinate-convention update further demonstrates why a small reprojection residual can accompany a substantially different physical mapping error.

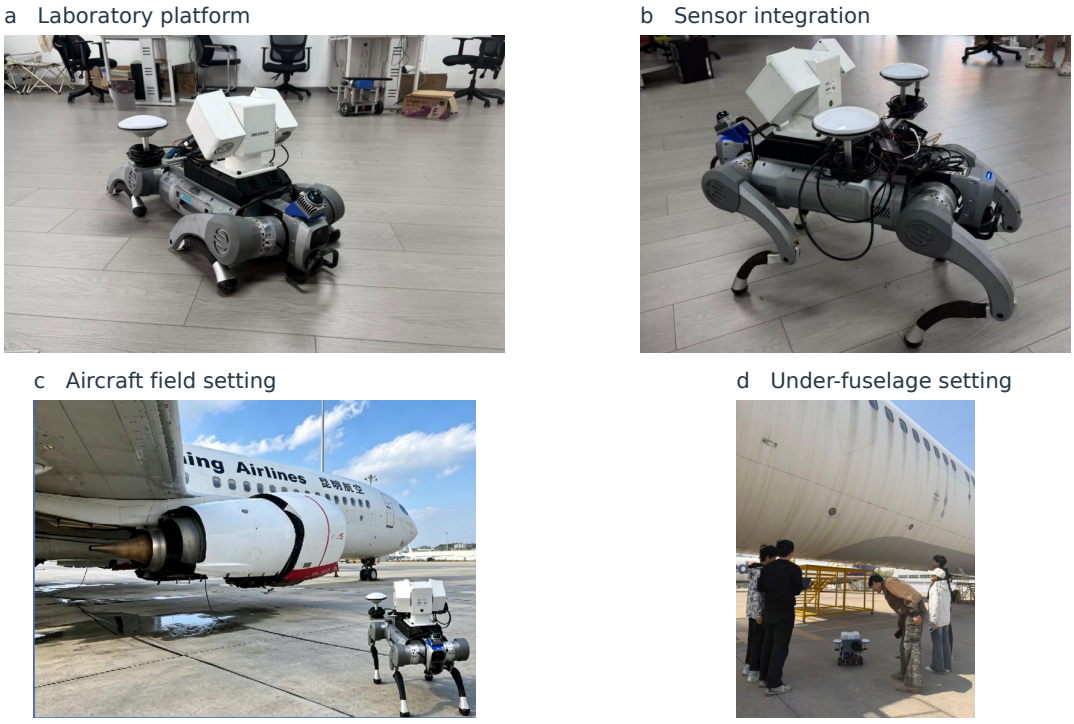


Figure 1: Project platform and aircraft setting: laboratory integration, sensor integration, aircraft field setting and under-fuselage setting. These original photographs locate the engineering workflow addressed by the localization, acquisition and CAD-association services.

2. Related work

2.1. Engineering knowledge and component changes

Process knowledge and decision knowledge graphs organize engineering choices and their supporting context [1, 2]. Semantic change propagation identifies component relations for impact analysis [3, 6]; inspection ontologies connect operating conditions to robot requirements [7]. Representational completeness and expressiveness [8], and executable model conversion [9], supply useful evaluation perspectives. Here the engineering operation is the applicability of an existing statistical statement after a request or provider changes. It requires links between the requested quantities, their implemented dependencies, the original event and the population used for calibration.

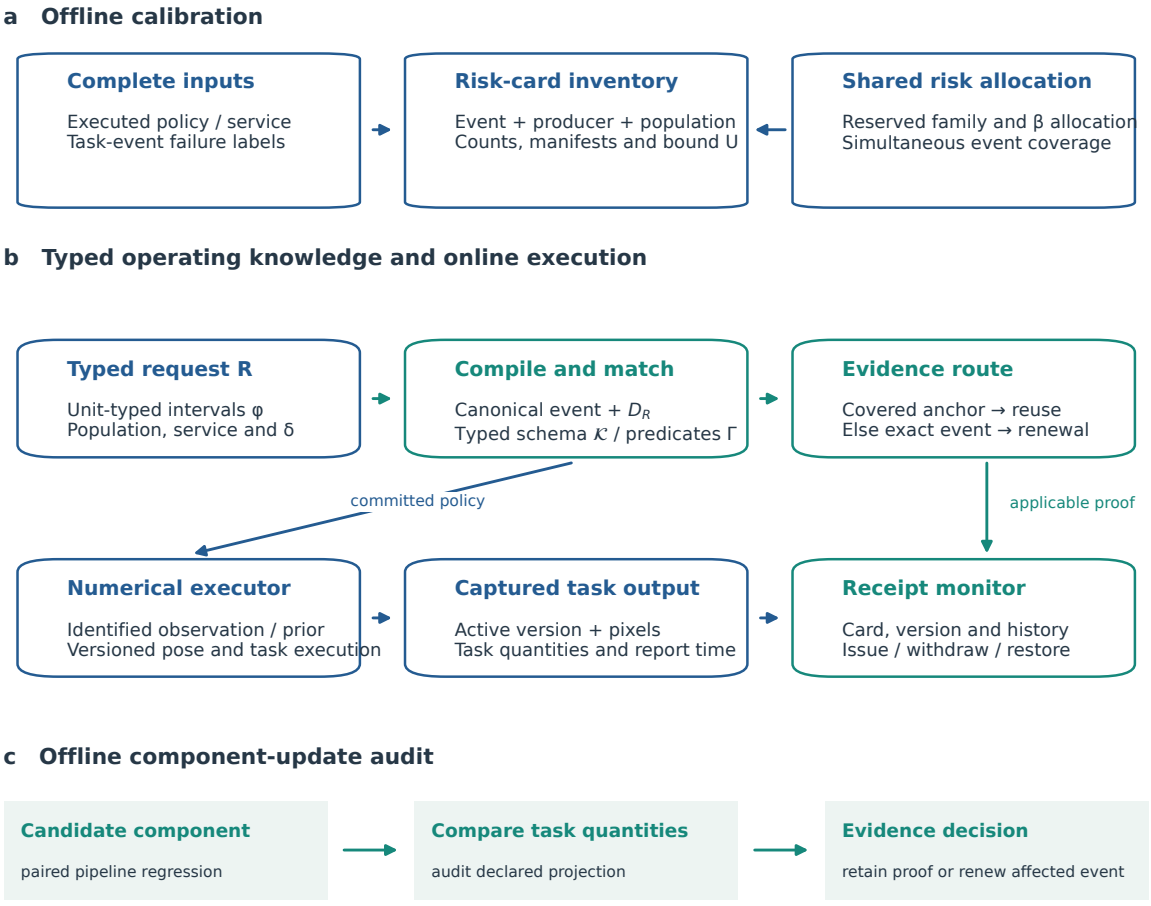


Figure 2: Executable evidence-management architecture. Calibration records establish event bounds; typed requests select compatible evidence and a producer plan. Numerical versions and execution history govern attachment. Component updates pass through regression and dependency checks before their evidence is retained or renewed.

2.2. Numerical services and versioned execution

Anytime computation and metareasoning connect numerical quality to processing cost [10, 11, 12, 13, 14]. Registration and state estimation provide the transformations used by a CAD-guided inspection workflow [15, 16, 17]; covariance studies characterize their uncertainty [18, 19, 20]. Our interface treats an estimate as a version of an identified source. This enables a statement to refer to the numerical transformation used by a specific image association. The image benchmark isolates this interface through prior-constrained pose refinement from pixel-derived fastener correspondences.

2.3. Calibrated risk and evidence lifetime

Assumption–guarantee contracts express component obligations [21]. Binomial inversion and risk-controlled selection establish finite-family guarantees for specified programs [22, 4, 23]. Runtime assurance monitors assumptions and switches admissible execution [5]. The operating contract connects these mechanisms to evidence issuance: containment identifies a stronger certified event, dependency projection identifies a compatible producer, and execution history identifies the output to which the statement can attach.

Table 1: Positioning by engineering operation.

Established operation	Executable relation	Evaluated endpoint
Knowledge and change reasoning	Requested quantities link component versions to certified events	Retained and renewed evidence after component updates
Numerical estimation	Source versions link a pose to its image-to-CAD mapping	Pixel residual, spatial error and trace identity
Risk-controlled selection	Containment links changed requirements to calibrated bounds	Supported contexts and constructed proof catalogue
Runtime monitoring	Persistent history links attachment attempts to issuance	Invalid-record refusal and restoration

3. Executable task and evidence model

3.1. Typed representation and compilation

The operating knowledge structure is

$$\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma), \tag{1}$$

where entities have types τ, typed relations $\mathcal{R}$, immutable specifications σ and executable predicates Γ. Requests require events; cards bound events and support plans; plans select policies; execution records use numerical versions. Quantities depend on providers. A source-revision relation requires the same source key and a strictly increasing version sequence. The schema connects an engineering requirement to the evidence and execution information needed to decide its applicability.

A request $R = (g, a, \varphi, \delta)$ fixes population, service, a conjunction of quantity intervals and a risk allowance. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi. \tag{2}$$

The compiler checks engineering dimensions, converts units, intersects repeated intervals and produces a canonical event identity. Omitted bounds give unbounded sides. Nonfinite required quantities fail the event. Compilation also aggregates the providers used by the requested quantities. The implemented manifest declares data-flow dependencies, while regression evaluates their numerical consequences.

For a card A, producer p and committed policy π, admission follows

$$\begin{aligned} &\mathrm{ValidCard}(A) \wedge \mathrm{Match}(A, R, \pi) \wedge \mathrm{Cover}(A, R) \\ &\quad \wedge \mathrm{SameProjection}_{D_R}(p, p_A) \wedge U_A \leq \delta \Longrightarrow \mathrm{Admissible}(\pi, A, R). \end{aligned} \tag{3}$$

Component revisions are checked by an offline regression gate. It compares outputs on common inputs with the proposed affected quantities; an undeclared difference produces a registration veto. Request processing then uses registered signatures, source versions and execution history.

Table 2: Typed operating records and their decisions.

Record	Required information	Executable operation
Request/event	Population, service, quantity units and bounds	Compile the failure event
Producer/quantity	Function version, provider dependencies	Compare requested signatures
Source/version	Observation key, sequence, payload integrity	Track the estimate used by the output
Card	Event, producer, population, calibration lineage, n, k, U	Check bound and containment
Plan/record	Request, policy, evidence, active source and history	Issue, withdraw or restore evidence

3.2. Pose refinement and image-to-CAD quality

The image producer receives an identified pose prior and an image observation. Pixel extraction finds fastener centers; the prior supplies coarse CAD-to-pixel predictions for correspondence assignment. Pose refinement uses these matched coordinates and the prior:

$$\widehat{\xi} = \arg\min_{\xi} \sum_{i\in\mathcal{I}} \sum_{d\in\{x,y\}} \rho\left(\frac{\pi_\xi(q_i)_d - u_{i,d}}{\sigma_u}\right) + \sum_{d=1}^{3} \left[\rho\left(\frac{\Delta t_d}{\sigma_t}\right) + \rho\left(\frac{\Delta\theta_d}{\sigma_\theta}\right)\right]. \tag{4}$$

Here q_i is a CAD address, u_i a pixel-derived center, and π_ξ the perspective projection. Translation Δt and local rotation $\Delta\theta$ perturb the supplied prior. The componentwise Huber loss has transition 2; σ_u, σ_t and σ_θ scale pixels, translation and rotation. Matching and estimation use the prior and image pixels. Known scene geometry provides the offline quality labels.

Reprojection and spatial mapping measure different properties. Reprojection compares the fitted prediction with the decoder's coordinate. Spatial error evaluates the image-to-CAD map $H_{\widehat{\xi}}$ at a fixed physical image location $u_i^\star$:

$$e_{\mathrm{CAD}} = \max_{i\in\mathcal{I}_c} \|H_{\widehat{\xi}}(u_i^\star) - q_i^\star\|. \tag{5}$$

A systematic pixel-coordinate shift can be absorbed by pose refinement. The fitted residual then describes agreement with the shifted decoder, while the mapping of the actual image location changes. This distinction gives the evidence model a concrete engineering role: a decoder version is part of the certified mapping producer.

An execution record binds the image, prior and refined pose to one source. New eligible estimates replace that source's active numerical version in order. Attachment checks the version used by the output and retains a persistent record of invalid attempts. The supplementary numerical study evaluates the same source/version semantics with retained scan registration and motion replay.

3.3. Calibration and persistent issuance

For a specified producer and event with k failures among n independent complete calibration inputs, the one-sided bound is

$$U(k,n;\alpha) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1-\alpha; k+1, n-k), & k < n, \end{cases} \qquad \alpha = \beta/K. \tag{6}$$

The benchmark uses $\beta = 0.05$ and $K = 6 \times 8 = 48$ producer/event combinations. The union bound gives one simultaneous confidence event for the complete fixed family. Calibration-dependent selection can use any bound in this family; held-out test observations characterize the selected services separately.

Runtime issuance checks source integrity, active version, requested dependency signatures and persistent history. If $A_{s,j}$ is the current attachment condition and $M_{s,j}$ is the source-history state,

$$M_{s,j} = M_{s,j-1} \wedge A_{s,j}, \qquad M_{s,0} = \text{true}. \tag{7}$$

Reconstructing a plan retains this state. A detected invalid attempt therefore prevents later attachment in the same execution scope while numerical processing continues through its original interface.

4. Evidence transfer through engineering changes

4.1. Requirement containment and provider projection

A requirement specifies acceptable quantity values and the corresponding failure event:

$$S_R = \{z : \ell_{R,j} \le z_j \le u_{R,j},\ j \in J_R\}, \quad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{8}$$

Table 3: Reference contract procedure.

Step	Operation
1	Compile requested quantities, engineering units, bounds and providers.
2	Match producer, population, policy and calibration lineage.
3	Retain a compatible covering card whose bound satisfies the allowance.
4	Construct an exact-event card for a remaining request; refuse if its bound fails.
5	Commit the evidence plan before evaluating the held-out input.
6	Track source versions, output identity and persistent execution history.
7	Issue, withdraw or restore the applicable statement.

An anchor covers the request when its required quantities include J_R and its intervals are at least as strict. Consequently,

$$S_A \subseteq S_R \Longrightarrow E_{p,R} \subseteq E_{p,A} \Longrightarrow \Pr(E_{p,R}) \leq U_{p,A}. \tag{9}$$

The entire containment closure is covered on the same confidence event. A relaxed blur or displacement requirement can therefore use a stronger existing statement.

For quantity j, let D_j contain its providers and $D_R = \bigcup_{j \in J_R} D_j$. The requested signature projects the producer snapshot onto D_R. Transfer after a provider change requires

$$z_{p',j}(X) = z_{p,j}(X) \quad (j \in J_R) \Longrightarrow E_{p',R} = E_{p,R}. \tag{10}$$

In the imaging producer, GSD follows range, focal length and incidence. Blur additionally follows exposure and motion. Coverage, pixel association and refined-pose mapping additionally depend on raster size, illumination and decoding. An ROI or illumination update can therefore retain optical evidence while renewing the affected image-to-CAD statements.

Theorem 1 (Coverage of an authorized event)**.** *For one fixed calibration family, let anchor A cover request R with $U_A \leq \delta$. Assume a matched population and invariant requested functions. On the simultaneous confidence event, $\Pr(E_R) \leq \delta$. For an issuance indicator I generated by the persistent-history monitor,*

$$\Pr(E_R \cap \{I = 1\}) \leq \delta. \tag{11}$$

An invalid attempt prevents subsequent issuance within that source scope.

Proof. Requested-function identity preserves the request event. Containment gives $E_R \subseteq E_A$, and simultaneous calibration gives $\Pr(E_A) \leq U_A \leq \delta$. Intersecting with issuance preserves the bound. The conjunction in Eq. (7) remains false after an invalid attempt. □

4.2. Reuse with exact-event fallback

For a committed producer/policy, let $\mathcal{A}_\pi(R)$ contain compatible covering anchors and $\mathcal{D}_\pi(R)$ contain compatible exact-event cards. The route is

$$Q_\pi(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}_\pi(R) \neq \varnothing, \\ \text{direct}(D), & \mathcal{A}_\pi(R) = \varnothing,\ \mathcal{D}_\pi(R) \neq \varnothing, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{12}$$

Both branches use the same fixed confidence allocation. An exact-event fallback preserves support when the available anchor is too strict. The algorithm separates this request decision from the producer computation required to renew its quantities.

4.3. Matched comparison methods

Whole-producer renewal reconstructs every requested event after a producer update. Dependency-aware exact renewal retains compatible exact-event cards but constructs each uncovered event separately. The contract additionally uses containment and exact fallback. A stale-cache control keeps the baseline producer's cards without checking applicability. All methods receive the same calibration records, quantities, family allocation, requirements and numerical producer. The comparison measures complete catalogue construction, including refusal decisions, and separately measures the logical source work for individual cold update/request cases.

5. Experimental design

5.1. Independent CAD-surface scenes

The benchmark generates planar, singly curved and doubly curved metallic surfaces with an equal-probability independent family draw. Principal curvature radii are 2.5–5.0 m and 4.0–8.0 m. Each surface has a planned 7×5 fastener grid with 35-mm spacing and 2.5-mm head radius. Perspective ray/surface intersection creates the raster. Surface-normal illumination, a specular term, brushed texture, panel seams, exposure filtering and intensity noise provide the observed pixels.

The surface simulation has 192 calibration and 192 held-out test scenes, with independently sampled camera pose, curvature, focal length, exposure, motion, illumination and prior error. Development inputs are excluded from both sets. Baseline ROI is 640×480, expanding to 896×672. Six producer configurations share each complete input, giving 2,304 executed producer/input combinations. The configuration changes focal length by 15%, exposure by 50%, raster size, illumination by a factor 0.45, or the decoder coordinate origin by 12 pixels.

Pixel extraction uses local contrast, connected components and intensity-weighted centers. CAD assignment uses the pose prior and a 30-pixel gate; robust nonlinear least squares refines the pose with 0.5-pixel observation scale, 3-mm translation prior and 0.001-rad rotation prior. The initial pose has a 2% additional 5-mm uncertainty component. Figure 3 shows actual inputs and pixel-derived outputs for the first test scene in each geometry family.

Table 4: Imaging and estimation operating envelope.

Quantity	Specification
Camera distance	0.70–1.15 m, with independently sampled lateral position
Pixel focal length	1,200–1,600 pixels
Exposure and motion	3–7 ms; 0.06–0.18 m/s
ROI	640×480 baseline; 896×672 expanded
Scene noise	1.5–3.0 intensity units, plus surface texture and lighting
Pose refinement	Robust pixel/CAD least squares with translation and rotation priors
Calibration/test split	192/192 complete independent scenes

5.2. Requests, calibration and measurement

Eight requests span optical, acquisition, association and complete service, each with a strict and relaxed specification. Optical bounds use GSD 1.2/1.5 mm per pixel and blur 3/4 pixels. Acquisition uses GSD 1.0/1.2 mm, blur 2/3 pixels and coverage 0.8/0.6. Association uses coverage 0.8/0.6, reprojection 8/12 pixels and spatial mapping 8/12 mm. Complete service combines acquisition and association. The four strict events supply the anchor catalogue.

Cards use only calibration outcomes. The primary allowance is 0.15, with $\beta = 0.05$ allocated over all 48 producer/event combinations. Held-out observations do not determine thresholds, producer definitions or admission. Quality scores distinguish fitted pixel residual from the mapping of a physical image location. All pose fitting precedes evaluation against scene truth.

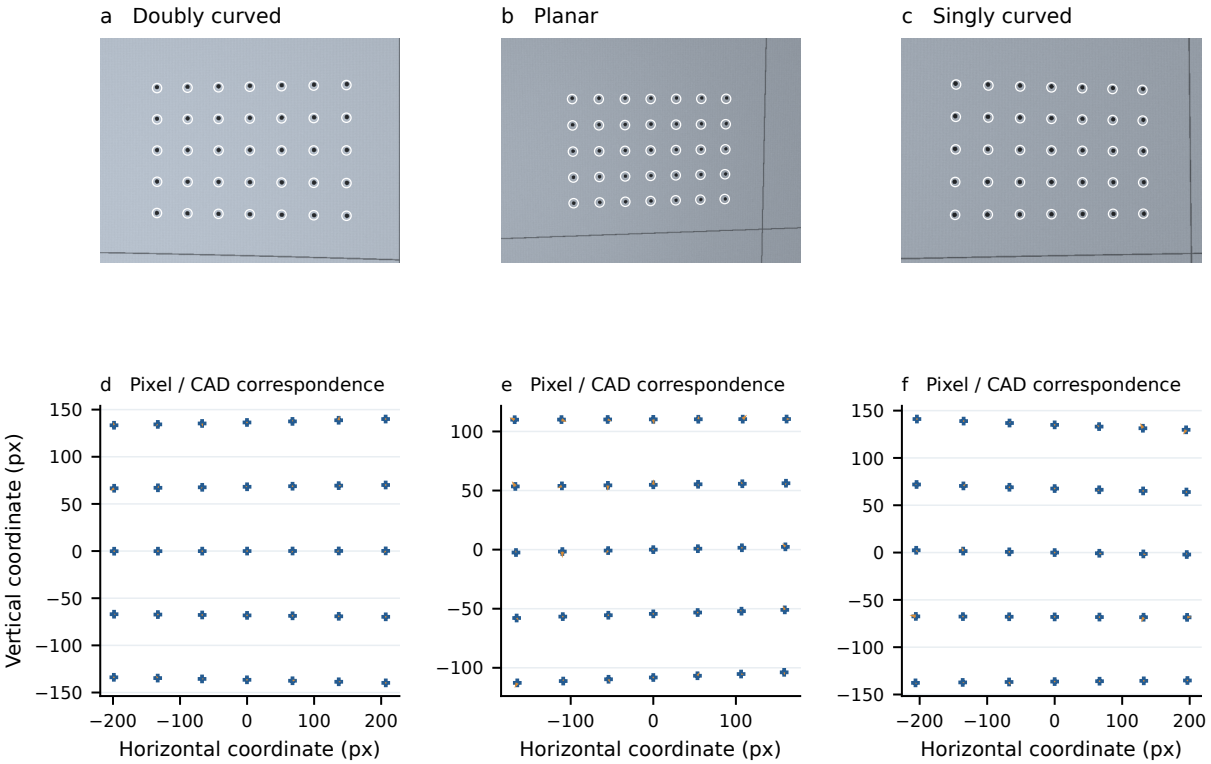


Figure 3: Independent CAD-surface inputs and outputs. Top: actual rendered images and pixel-derived fastener centers. Bottom: CAD-projected and decoded coordinates, with residual arrows magnified for visibility. The displayed scenes are the first held-out input in each geometry family.

Complete-catalogue construction is measured with identical stored calibration quantities and a shared confidence allocation. Counts include every card constructed to reach an admission or refusal. CPU medians use 11 repetitions. A separate ledger evaluates 40 individual cold cases: five updates times eight requests, each with a 192-input renewal block. This ledger distinguishes full image pipelines, optical-only evaluations and retained inputs; shared catalogue batching is a separate workload. Execution uses Linux/Python 3.12, one numerical thread per worker, and an AMD EPYC 9V74 host. Complete request specifications, per-scene outputs and provider definitions are given in the Supplementary Material.

6. Results

6.1. Pixel fitting and physical mapping

On the 192 held-out baseline scenes, mean correct-target coverage is 99.96%. Pixel-based refinement reduces median translation error from 4.392 to 0.997 mm. The baseline median reprojection residual is 0.217 pixel; median and 95th-percentile physical mapping errors are 0.056 and 0.115 mm.

The decoder-coordinate update changes the result in a different direction. Median reprojection remains 0.261 pixel, while median physical mapping error rises to 8.217 mm and its 95th percentile to 10.673 mm (Fig. 4). The estimator fits the new coordinates, but the transformation of actual image locations is displaced. The observed residual therefore does not identify the validity of the original mapping evidence.

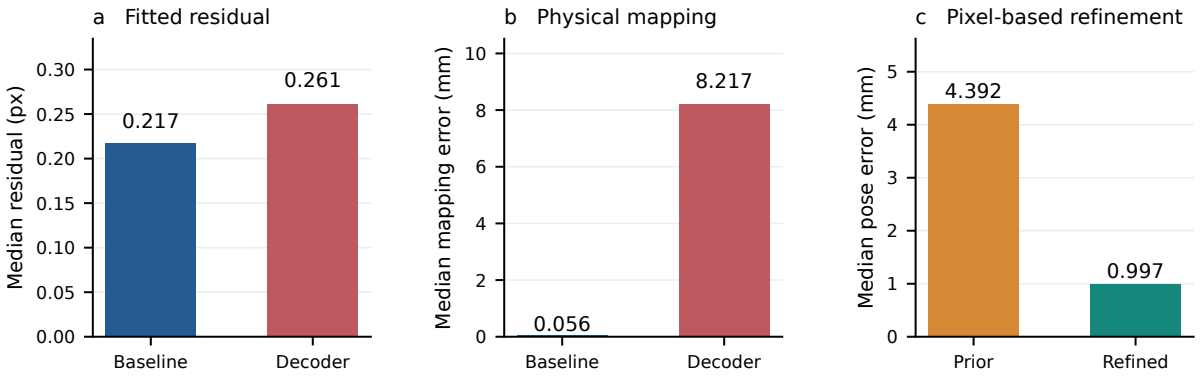


Figure 4: Pixel fitting and physical mapping on held-out scenes. A decoder-coordinate shift changes spatial mapping while leaving a small fitted residual. Baseline image correspondences refine the supplied pose prior.

6.2. Calibrated support across producer changes

At allowance 0.15, the fresh catalogue supports 44 of the 48 producer/request contexts. Exposure growth refuses strict acquisition and complete service. The decoder update refuses strict association and complete service. Their relaxed requirements remain supported. Figure 5 shows all bounds, including refusals.

For strict acquisition, the exposure update increases calibration failures from 3/192 to 22/192 and the bound from 0.066 to 0.201. The held-out failure count is 34/192 (17.71%). For strict association, the decoder update gives 110/192 calibration failures, bound 0.681, and 106/192 held-out failures (55.21%). These results connect evidence withdrawal to observed changes in the complete producer.

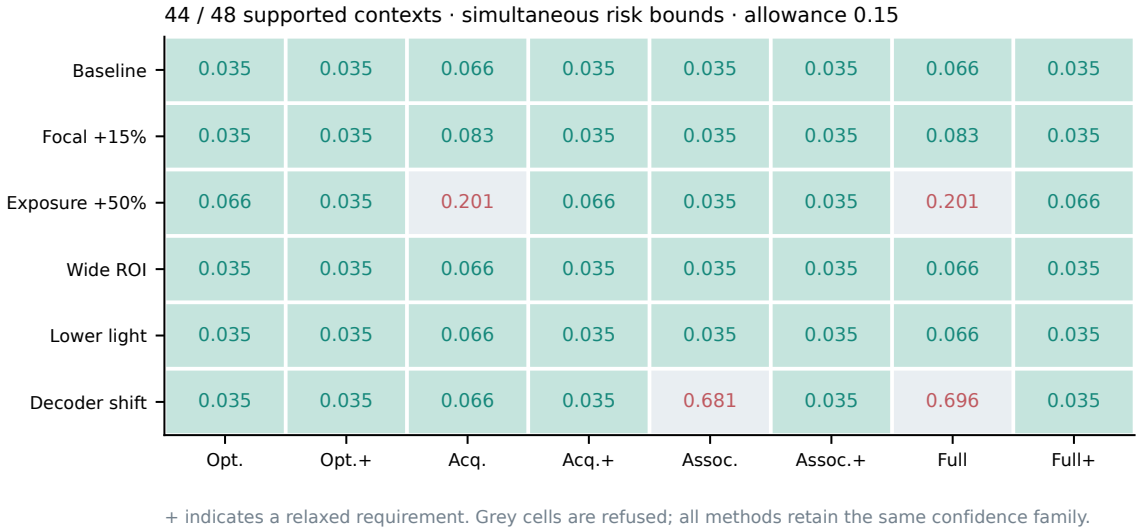


Figure 5: Complete producer/request catalogue. Every cell displays its simultaneous upper risk bound. Green cells satisfy allowance 0.15; grey cells are refused. A plus sign denotes a relaxed requirement.

6.3. Complete evidence construction and fallback

All three matched renewal methods support the same 44 contexts. Whole-producer renewal constructs 48 cards, dependency-aware exact renewal constructs 42, and the contract constructs 25, including cards needed for refusal decisions and four relaxed-event fallbacks. The corresponding calibration-record evaluations are 9,216, 8,064 and 4,800: a 40.48% reduction relative to dependency-aware exact renewal and 47.92% relative to whole-producer renewal.

The stale-cache control issues statements in all 48 contexts, including the four contexts unsupported by the current catalogue. Its higher issuance count therefore reflects retained baseline statements, rather than added current-producer evidence. The fallback route preserves directly supportable requests when a strict anchor cannot authorize them.

Table 5: Complete catalogue construction at the same confidence allocation. Counts include refusal decisions.

Method	Supported	Constructed cards	Record evaluations	CPU (ms)
Whole-producer renewal	44/48	48	9,216	6.720
Dependency-aware exact	44/48	42	8,064	6.332
Contract + fallback	44/48	25	4,800	4.723

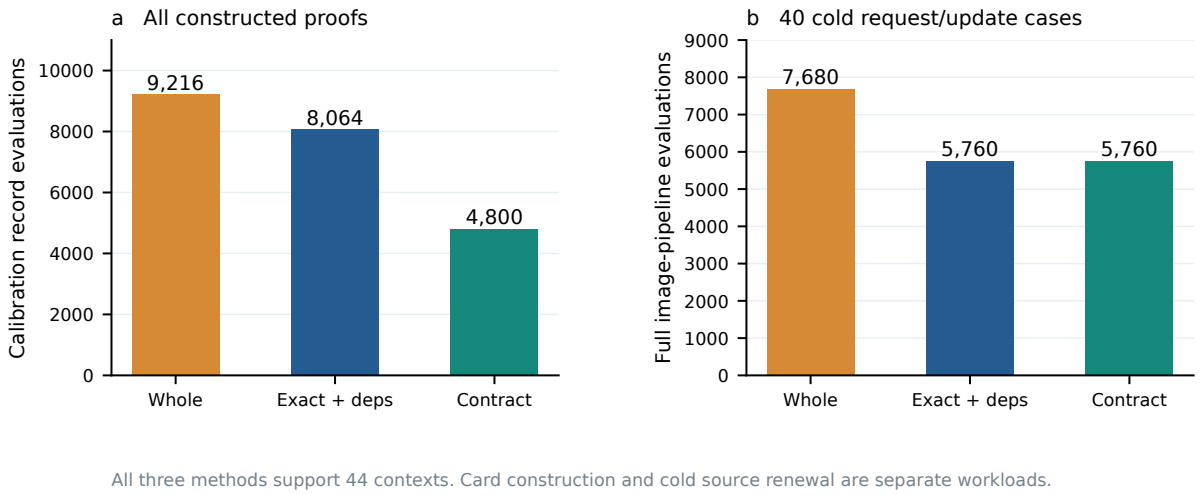


Figure 6: Evidence construction and source renewal. (a) Actual record evaluations for complete catalogues, including refusals. (b) Logical full-pipeline evaluations across 40 independent cold update/request cases. Both views retain the same support; their measurement boundaries are distinct.

6.4. Requested-function renewal and recovery

Across the 40 cold update/request cases, whole-producer renewal requires 7,680 full-pipeline evaluations. Requested dependency projection reduces this to 5,760, with 768 optical-only evaluations and 1,152 retained inputs. The optical-only implementation is checked numerically against the full producer on all 1,152 calibration/configuration combinations. No image generation or decoding is required for that branch.

The catalogue workload batches shared producer quantities before constructing cards; the cold ledger instead evaluates one independently initialized request per update. Producer matching determines the needed stages, while containment determines the needed event statements. Figure 7 shows how an exposure change preserves GSD but alters the acquisition bound. Restoring the original producer signature restores the corresponding original evidence lineage.

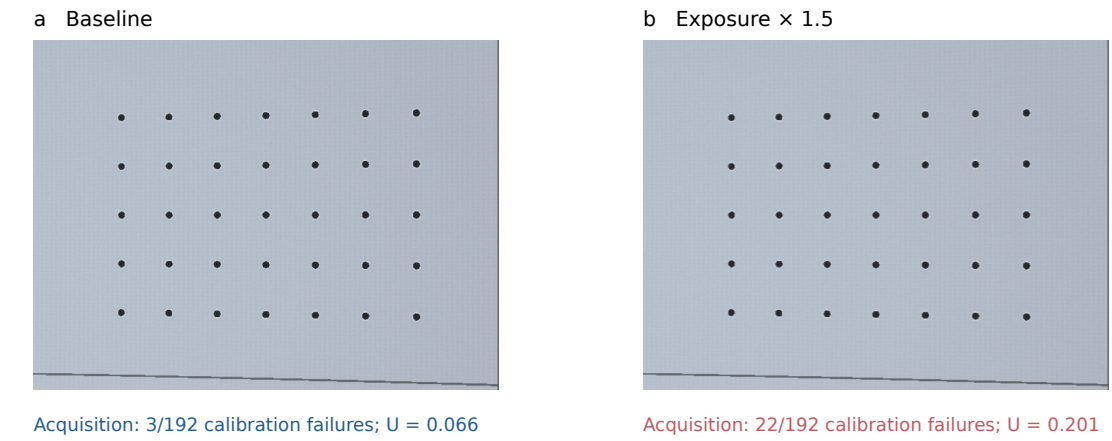


Figure 7: Actual images and population evidence under an exposure update. The displayed input is fixed. Optical sampling is retained while blur and the strict acquisition bound change. A relaxed request remains supported through exact-event fallback.

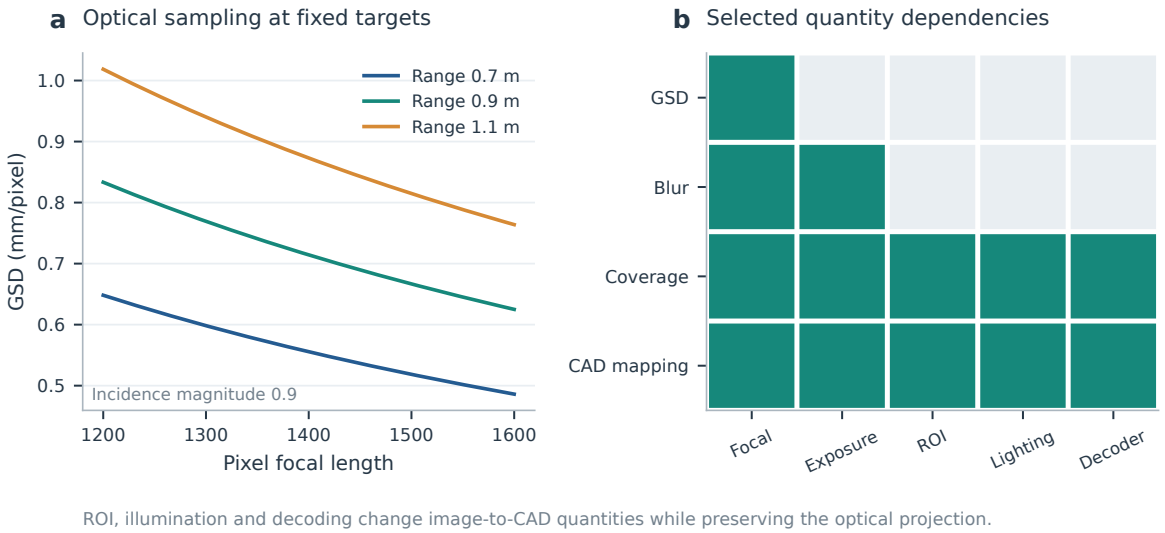


Figure 8: Optical quantities and selected dependencies. GSD follows range, focal length and incidence. ROI, illumination and decoding act through acquired pixels and mapping while preserving the optical projection under their registered update definitions.

6.5. Confidence and allowance sensitivity

Sensitivity varies confidence error and risk allowance over the fixed 48-card family (Fig. 9). The primary allocation and all primary decisions remain unchanged. The held-out acquisition and association failures identify which component changes create the elevated risk. Requirement relaxation supplies additional support through its event definition, while a tighter confidence allocation changes the admission margin.

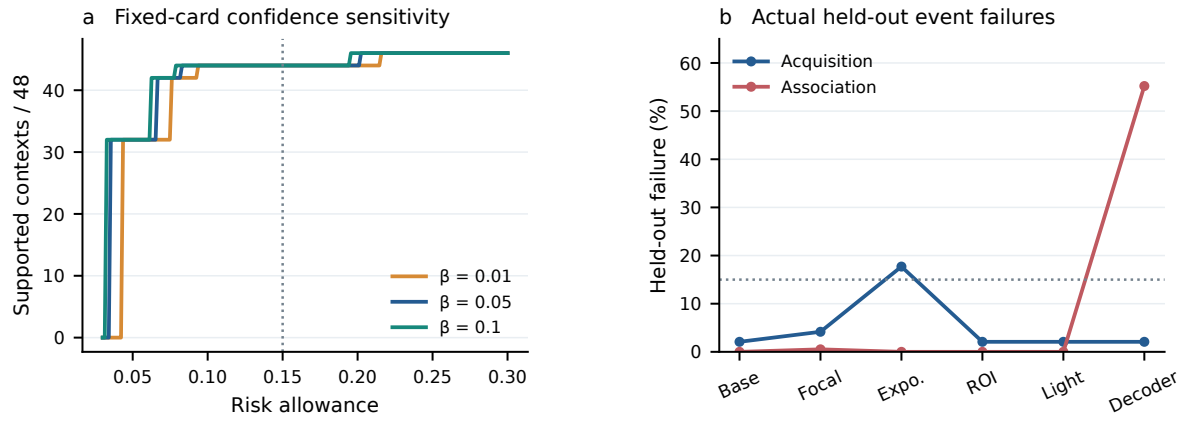


Figure 9: Fixed-family sensitivity and held-out event failures. The primary design uses confidence error 0.05 and allowance 0.15. The sensitivity curves use calibration counts; held-out event rates are reported separately.

7. Discussion

The physical meaning of a quality quantity determines the evidence it needs. The coordinate-origin experiment illustrates the distinction: pose refinement can explain shifted decoder coordinates with a small residual, while the mapping of actual image pixels changes. A statistical statement for the original producer consequently carries information that a current residual alone does not supply. Binding that statement to the event, producer and numerical source makes the change visible to the information system.

Requirement change and producer change also have different consequences. A relaxed threshold changes the accepted event; containment supplies a valid stronger bound. A changed decoder alters the function supplying the quantity; a matching calibration lineage or a valid requested-function transfer is then required. The exact fallback preserves requests that remain supportable after a strict anchor fails. These operations explain the shared 44-context support and the smaller constructed catalogue.

The workload decomposition connects representation to engineering action. Dependency projection identifies inputs whose requested quantities remain unchanged and identifies the producer stages required by an affected request. Containment then reduces the event catalogue constructed from those quantities. The same model can therefore determine both what must be recomputed and which existing statistical statement can be retained. The image/CAD benchmark provides executable instances of those decisions, while the supplementary source/version studies evaluate attachment history and restoration.

8. Conclusion

Executable operating contracts connect CAD-guided inspection requirements, versioned numerical producers and calibrated evidence. Pixel-based estimation demonstrates why a small fitted residual can accompany a changed spatial mapping. Reuse with exact fallback preserves all 44 supported contexts while reducing complete-catalogue record evaluations by 40.48%. The framework makes requirement, component and execution changes actionable through selective evidence retention, renewal and issuance.

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